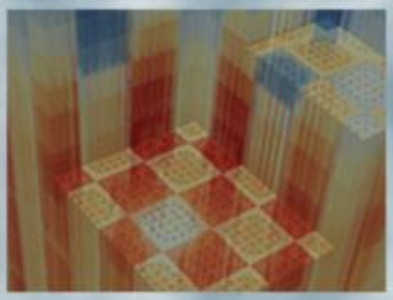
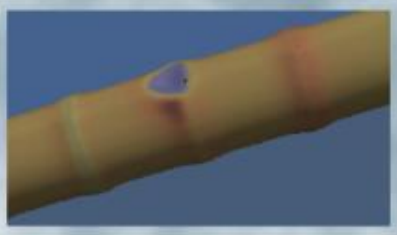
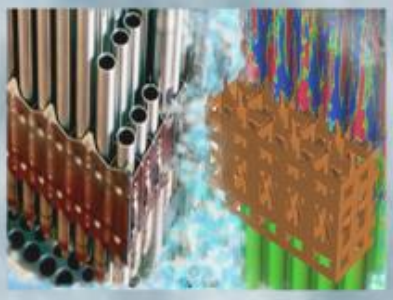
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Consortium for Advanced Simulation of LWRs

VERAIn Software Requirements, Test Plan, and Test Report

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10/1/2020

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Revision Log

Revision	Date	Affected Pages	Revision Description
0	7/31/2019	All	Initial Release
1	11/15/2019	All	4.1 Release
2	10/1/2020	All	4.2 Release

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VERAIn Software Requirements, Test Plan, and Test Report

Approvals:

Erik Walker

Digitally signed by Erik Walker
Date: 2021.05.13 19:01:38 -04'00'

5/13/2021

Erik Walker, VERAIn Product Software Manager **Date**

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Date: 2021.05.14 13:36:41 -04'00'

5/14/2021

Independent Reviewer **Date**

Executive Summary

This document describes the software test plan for VERAIn and provides appendices for the software requirements and software test report. In this document, the test platform hardware and software are described. Section A provides a list of the tests run and their acceptability as a test report. Section B provides the list of low-level software requirements as a requirements traceability matrix.

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1. Testing Procedure

1.1 Computer Program Tested

1.1.1 Program: VERAIn

Built under VERA Environment with following packages and versions:

Listing 1.1: Software Versions Evaluated

```
*** Base Git Repo: VERA
4dd64f9 [Wed Aug 5 09:43:50 2020 0400] <grahamam@ornl.gov>
Merge branch 'set_MPI_EXEC_MAX_NUMPROCS' into 'master'
*** Git Repo: TriBITS
91aaa7a [Mon Nov 11 14:33:26 2019 0500] <bairdml@ornl.gov>
Merge branch '6124 CheckinTest update' into 'master'
*** Git Repo: Trilinos
3b9c7e3 [Thu Jan 30 00:18:02 2020 0500] <collinsbs@ornl.gov>
Merge pull request #1 from bartlettroscoe/casl phi 6249 upgrade tribits trilinos
*** Git Repo: TeuchosWrappersExt
781aae8 [Fri May 10 09:51:36 2019 0600] <rabartl@sandia.gov>
Add missing std include (PHI 5294)
*** Git Repo: Futility
6739369 [Thu Jul 30 20:00:39 2020 0400] <collinsbs@ornl.gov>
Merge branch 'master' of https://github.com/CASL/Futility
*** Git Repo: MAMBA
fac6b4a [Sun Aug 16 04:00:35 2020 0400] <collinsbs@ornl.gov>
Merge branch 'master' of code int.ornl.gov:mamba/MAMBA
*** Git Repo: VERAInExt
cb78eef [Fri Aug 21 00:20:49 2020 0400] <walkered@ornl.gov>
Merge branch 'fix_manual_errors' into 'master'
*** Git Repo: DataTransferKit
19526ab [Fri Aug 18 09:14:16 2017 0400] <dalg24@gmail.com>
Merge pull request #297 from naughtont3/tjn fortran off
*** Git Repo: COBRA TF
854bf00 [Thu Aug 20 20:00:40 2020 0400] <collinsbs@ornl.gov>
Merge branch 'master' of code int.ornl.gov:rsk/CTF
```

```
*** Git Repo: VERAData
a81122d [Tue Jul 2 16:05:52 2019 0400] <stimpsonsg@ornl.gov>
Merge branch 'endf8.0_mpact 51g_origenlibrary' into 'master'
*** Git Repo: VERAData/CEData
8252504 [Wed Jul 8 20:44:25 2020 0400] <pandyatm@ornl.gov>
Merge branch 'update hydrogen pole data' into 'master'
*** Git Repo: MOOSEExt
f721711 [Thu Jun 11 18:39:51 2020 0400] <stimpsonsg@ornl.gov>
Merge branch 'decay heat' into 'master'
*** Git Repo: MOOSEExt/MOOSE
4dc6cda [Tue Oct 1 15:20:45 2019 0400] <stimpsonsg@ornl.gov>
Merge branch 'merge_sync' into 'master'
*** Git Repo: VERAOneWay
56037bf [Mon May 25 00:14:09 2020 0400] <collinsbs@ornl.gov>
Adds gitlab ci file
*** Git Repo: SCALE
90b01c0 [Wed Aug 19 20:42:22 2020 0400] <pandyatm@ornl.gov>
Merge branch 'sync scale' into 'master'
*** Git Repo: XSTools
96c2e33 [Sat Aug 15 00:24:09 2020 0400] <collinsbs@ornl.gov>
Merge branch 'perturb_gamma' into 'master'
*** Git Repo: MPACT
7b69e67 [Fri Aug 21 20:01:11 2020 0400] <collinsbs@ornl.gov>
Merge branch 'master' of code int.ornl.gov:MPACT/MPACT
*** Git Repo: VeraShift
90800d3 [Fri Aug 14 12:07:38 2020 0400] <pandyatm@ornl.gov>
Merge branch 'sync verashift' into 'master'
*** Git Repo: DakotaExt
afd354f [Tue Aug 1 19:36:55 2017 0400] <rhoope@sandia.gov>
Fix tarball installation
*** Git Repo: DakotaExt/Dakota
7415238 [Thu Jun 15 12:40:49 2017 0400] <rhoope@sandia.gov>
Triage static builds of Dakota 6.6
*** Git Repo: VUQDemos
b2d0175 [Tue Oct 8 14:13:19 2019 0400] <salkork@ornl.gov>
Merge branch '1 correcting hardcoded python paths in vuq core' into 'master'
*** Git Repo: VERAView
28e0f8a [Thu Mar 31 16:53:16 2016 0400] <leerw@ornl.gov>
Sync from code.ornl.gov for version 1.0
```

VERAIn has no dependencies on any other code package contained within VERA.

1.1.2 System Software

The operating system on which these tests were performed was Linux 3.10.0-1127.18.2.el7.x86_64 .

1.1.3 Test Calibration

None.

1.1.4 Computer Hardware Used for Testing

rnsdinf01.ornl.gov

1.2 Test Equipment Calibration

None.

1.3 Date of Test

Started: Aug. 23, 2020 - 05:40:32 UTC
Total Test Time: 1 minute, 56 seconds

1.4 Data Recorder

ctest-3.14.3 and CDash Version trilinos-release-2018-08-prebuilt

1.5 Simulation Models Used

Since VERAIn is only an input parser, it contains no simulation models.

1.6 Test Problems

A list of test problems and their requirements may be found in Appendix B of this document.

1.7 Results and Applicability

Results

Results are given in Appendix A.

Applicability

The test results from Appendix A satisfy the defined software requirements.

1.8 Action Taken in Connection with Noted Deviations

There were no noted deviations. Therefore, no actions were taken.

1.8.1 Person Evaluating Results

Erik Walker

1.9 Acceptability

The acceptance criterion is that all tests pass their individual criteria. As evidenced in column 4 of the table in Appendix A, all tests passed their criteria. The primary type of tests performed in the VERAIn test matrix are regression tests that compare a test output file to a version-controlled reference file that is deemed to be correct.

Regression test reference files are XML.gold files that have been generated and reviewed for correctness. Once the VERAIn parser is run on an input, the resulting XML file is compared to the corresponding XML.gold file using a straight diff. If the files are identical, the test passes. If any differences exist in the files, the test fails.

A few tests are designed to intentionally fail to capture common input mistakes and to ensure that the parser fails hard when these mistakes are encountered.

The test results were reviewed by the Product Support Manager and the Independent Technical Reviewer, who have determined that the computer program is acceptable for integration with the VERA Product Suite. The test results have demonstrated that the computer program(s) adequately and correctly performed. The testing covered the specified software requirements, proper handling of abnormal conditions or events, and interfaces with other components. The testing verified that the computer program did not introduce unintended consequences which degraded the overall software function. No discrepancies between expected and observed software functioning were identified.

2. Applicable Standards and Procedures

- NQA-1-2008 with NQA-1a-2009 Addenda [1]
- VERA-QA-004, VERA Software Configuration and Control
- VERA-QA-006, VERA Software Requirements, Design, and Testing

3. Required Records

3.1 CDash Test Results

See Section A.

3.2 Requirements Traceability Matrix

See Section B.

A. VERAIn Test Report

The following table was obtained from a CDash report. CDash is a web application included with CMake and CTest. CMake is the VERA build system, and CTest is a program that automatically runs the tests defined with CMake and sends results to CDash. All tests are included in the Git repositories as noted in List 1.1. Section B provides a list of the test names, along with the test input files and requirements.

Table 1: VERAIn Test Report from CDash

No.	Test Name	Details	Status
1	VERAIn_react2xml_bwr-peach-1	Completed	Passed
2	VERAIn_react2xml_bwr-peach-2	Completed	Passed
3	VERAIn_react2xml_bwr-peach-4	Completed	Passed
4	VERAIn_react2xml_bwr-plr	Completed	Passed
5	VERAIn_react2xml_bwr-thickthincb	Completed	Passed
6	VERAIn_react2xml_bwr-ge9	Completed	Passed
7	VERAIn_react2xml_bwr-peach-6	Completed	Passed
8	VERAIn_react2xml_bwr-peach-5	Completed	Passed
9	VERAIn_react2xml_bwr-axialvoid	Completed	Passed
10	VERAIn_react2xml_bwr-p6-test	Completed	Passed
11	VERAIn_react2xml_bwr-azixs	Completed	Passed
12	VERAIn_react2xml_bwr-peach-3	Completed	Passed
13	VERAIn_react2xml_bwr_jagged	Completed	Passed
14	VERAIn_react2xml_chen_3x3	Completed	Passed
15	VERAIn_react2xml_ex7_bwr	Completed	Passed
16	VERAIn_react2xml_apr1400-cyc1	Completed	Passed
17	VERAIn_react2xml_cross_w_baffle	Completed	Passed
18	VERAIn_react2xml_FAST_input_block	Completed	Passed
19	VERAIn_react2xml_ctf_convergence_terms	Completed	Passed
20	VERAIn_react2xml_fuel_options	Completed	Passed
21	VERAIn_react2xml_mox1	Completed	Passed
22	VERAIn_react2xml_MSED	Completed	Passed
23	VERAIn_react2xml_fast_3x3rod_HFP	Completed	Passed
24	VERAIn_react2xml_default_density_change	Completed	Passed
25	VERAIn_react2xml_pwr-CE-2a	Completed	Passed
26	VERAIn_react2xml_p6a_sa_1300_vuq	Completed	Passed

No.	Test Name	Details	Status
27	VERAIn_react2xml_par_cross	Completed	Passed
28	VERAIn_react2xml_pwr-CE-2b	Completed	Passed
29	VERAIn_react2xml_mamba_mass_bal	Completed	Passed
30	VERAIn_react2xml_pwr_CE_single_16x16	Completed	Passed
31	VERAIn_react2xml_misc_options	Completed	Passed
32	VERAIn_react2xml_Sample_3x3	Completed	Passed
33	VERAIn_react2xml_Sample_3x3_multistate	Completed	Passed
34	VERAIn_react2xml_Sample_7x7	Completed	Passed
35	VERAIn_react2xml_Sample_full_core_maps	Completed	Passed
36	VERAIn_react2xml_Sample_p3	Completed	Passed
37	VERAIn_react2xml_Sample_full_core_matmix	Completed	Passed
38	VERAIn_react2xml_Sample_full_core_wsym	Completed	Passed
39	VERAIn_react2xml_Sample_full_core_fuelmix	Completed	Passed
40	VERAIn_react2xml_Sample_full_core_reflector	Completed	Passed
41	VERAIn_react2xml_Sample_full_core_wsym_assm_even	Completed	Passed
42	VERAIn_react2xml_Sample_full_core	Completed	Passed
43	VERAIn_react2xml_Sample_full_core_wsym_state_assm	Completed	Passed
44	VERAIn_react2xml_Sample_full_core_wsym_state_assm_matopt	Completed	Passed
45	VERAIn_react2xml_Sample_full_core_wsym_state_dsym	Completed	Passed
46	VERAIn_react2xml_Sample_full_core_wsym_even	Completed	Passed
47	VERAIn_react2xml_Sample_p3_mpact	Completed	Passed
48	VERAIn_react2xml_Sample_full_core_wsym_state	Completed	Passed
49	VERAIn_react2xml_Sample_p3_unit_test	Completed	Passed
50	VERAIn_react2xml_Sample_p3_Anderson	Completed	Passed
51	VERAIn_react2xml_Sample_p4a_unit_test	Completed	Passed
52	VERAIn_react2xml_Sample_p4	Completed	Passed
53	VERAIn_react2xml_Sample_p6a_multifuel	Completed	Passed
54	VERAIn_react2xml_Sample_p6a	Completed	Passed
55	VERAIn_react2xml_Seabrook_5x5rod_HFP	Completed	Passed
56	VERAIn_react2xml_Sample_p4_mpact	Completed	Passed
57	VERAIn_react2xml_Sample_p7	Completed	Passed
58	VERAIn_react2xml_small_3x3rod_Bison	Completed	Passed
59	VERAIn_react2xml_Sample_qtr_core_wsym_3x3_standalone	Completed	Passed
60	VERAIn_react2xml_small_3x3rod_Bison_standalone	Completed	Passed
61	VERAIn_react2xml_small_3x3rod_BProd	Completed	Passed
62	VERAIn_react2xml_Sample_p9_jumpin	Completed	Passed
63	VERAIn_react2xml_Sample_p9_shuffle	Completed	Passed
64	VERAIn_react2xml_Sample_p9_restart	Completed	Passed
65	VERAIn_react2xml_small_3x3rod_HFP_2fuel	Completed	Passed
66	VERAIn_react2xml_small_3x3rod_HFP_8g	Completed	Passed
67	VERAIn_react2xml_small_3x3rod_HFP	Completed	Passed
68	VERAIn_react2xml_small_3x3rod_HZP	Completed	Passed
69	VERAIn_react2xml_small_5cross_assem_3x3rod_HFP	Completed	Passed
70	VERAIn_react2xml_small_3x3rod_IFBA	Completed	Passed
71	VERAIn_react2xml_small_multibundle_par	Completed	Passed

No.	Test Name	Details	Status
72	VERAIn_react2xml_small_qtr_core_3x3rod_HFP	Completed	Passed
73	VERAIn_react2xml_small_multibundle	Completed	Passed
74	VERAIn_react2xml_small_singlerod_HFP	Completed	Passed
75	VERAIn_react2xml_small_singlerod_HFP_8gb	Completed	Passed
76	VERAIn_react2xml_small_singlerod_HFP_8g	Completed	Passed
77	VERAIn_react2xml_disable_xml2ctf	Completed	Passed
78	VERAIn_react2xml_test_cobra_edits_inimat	Completed	Passed
79	VERAIn_react2xml_test_cobra_edits	Completed	Passed
80	VERAIn_veradiff_match	Completed	Passed
81	VERAIn_veradiff_fail	Completed	Passed
82	VERAIn_react2xml_smr-7x7	Completed	Passed
83	VERAIn_react2xml_decreasing-axial	Completed	Passed
84	VERAIn_react2xml_multi_scalar_error	Completed	Passed
85	VERAIn_react2xml_front-dash	Completed	Passed
86	VERAIn_react2xml_test_verashift_small_core_bioshield_no_pv	Completed	Passed
87	VERAIn_react2xml_test_verashift_3a	Completed	Passed
88	VERAIn_react2xml_test_verashift_3a_forward	Completed	Passed
89	VERAIn_react2xml_test-short	Completed	Passed
90	VERAIn_react2xml_test_verashift_3a_cadis	Completed	Passed
91	VERAIn_react2xml_test_verashift_4mini_nonunique	Completed	Passed
92	VERAIn_react2xml_with-dash	Completed	Passed
93	VERAIn_react2xml_2a_LS	Completed	Passed
94	VERAIn_react2xml_2a_LS_default	Completed	Passed
95	VERAIn_react2xml_test_verashift_4mini_nonunique_bioshield	Completed	Passed
96	VERAIn_xmlthexp_Sample_p6a	Completed	Passed
97	VERAIn_react2xml_test_verashift_small_core_bioshield	Completed	Passed
98	VERAIn_testVERAInMan	Completed	Passed

B. Requirements and Test Traceability Matrix

For the entries in the *Additional Info* field, the #NNNN is the CASL PHI Kanban ticket number with a hyperlink to this webpage.

Table 2: Requirements

Req. ID	*Requirement Description* *Test Name* *Test Input* *Additional Info*
1	The VERAIn package shall be able to parse VERA Progression Problem 2A with specified linear source options verain/test/2a_LS.inp None
2	The VERAIn package shall be able to parse VERA Progression Problem 2A with default linear source options verain/test/2a_LS_default.inp None
3	The VERAIn package shall be able to parse all FAST input options correctly verain/test/FAST_input_block.inp None
4	The VERAIn package shall be able to parse inputs in which multigrid solver options are specified verain/test/MSED.inp None
5	The VERAIn package shall be able to parse a full 3D core with an even core size verain/test/S_even.inp None

Req. ID	*Requirement Description* *Test Name* *Test Input* *Additional Info*
6	The VERAIn package shall be able to parse 9 3D 17x17 assemblies with feedback enabled in full symmetry verain/test/Sample_3x3.inp None
7	The VERAIn package shall be able to parse multiple state blocks containing differing options selected for the tinlet card and table input for inlet temperature as a function of power verain/test/Sample_3x3_multistate.inp None
8	The VERAIn package shall be able to parse 49 3D 17x17 assemblies with feedback enabled in full symmetry verain/test/Sample_7x7.inp None
9	The VERAIn package shall be able to parse a full 3D core verain/test/Sample_full_core.inp None
10	The VERAIn package shall be able to parse a full 3D core with fuel mixing verain/test/Sample_full_core_fuelmix.inp None
11	The VERAIn package shall be able to parse a full 3D core with a combination of full, quarter, and octant symmetric maps verain/test/Sample_full_core_maps.inp None

Req. ID	*Requirement Description* *Test Name* *Test Input* *Additional Info*
12	The VERAIn package shall be able to parse a full 3D core with material mixing. verain/test/Sample_full_core_matmix.inp None
13	The VERAIn package shall be able to parse a full 3D core with upper and lower reflector regions verain/test/Sample_full_core_reflector.inp None
14	The VERAIn package shall be able to parse a full 3D core with numerous BISON parameters enabled verain/test/Sample_full_core_wsym.inp None
15	The VERAIn package shall be able to parse a full 3D core with 18x18 assemblies. verain/test/Sample_full_core_wsym_assm_even.inp None
16	The VERAIn package shall be able to parse a full 3D core of size 16 with numerous BISON parameters enabled verain/test/Sample_full_core_wsym_even.inp None
17	The VERAIn package shall be able to parse a full 3D core with numerous BISON parameters enabled with multiple states verain/test/Sample_full_core_wsym_state.inp None

Req. ID	*Requirement Description* *Test Name* *Test Input* *Additional Info*
18	The VERAIn package shall be able to parse a full 3D core with numerous BISON parameters enabled with multiple states where the assemblies are defined over multiple ASSEMBLY blocks rather than all being defined in the same ASSEMBLY block verain/test/Sample_full_core_wsym_state_assm.inp None
19	The VERAIn package shall be able to parse a full 3D core with numerous BISON parameters including material thermal expansion options verain/test/Sample_full_core_wsym_state_assm_matopt.inp None
20	The VERAIn package shall be able to parse a full 3D core with numerous BISON parameters enabled with multiple states where one of the lattice definitions is twice as large as the others verain/test/Sample_full_core_wsym_state_dsym.inp None
21	The VERAIn package shall be able to parse Problem 3, which is a single 3D 17x17 assembly at BOL without feedback verain/test/Sample_p3.inp None
22	The VERAIn package shall be able to parse nonlinear Anderson Acceleration input options verain/test/Sample_p3_Anderson.inp None
23	The VERAIn package shall be able to parse Problem 3, which is a single 3D 17x17 assembly at BOL without feedback with numerous MPACT options verain/test/Sample_p3_mpact.inp None

Req. ID	*Requirement Description* *Test Name* *Test Input* *Additional Info*
24	The VERAIn package shall be able to parse Problem 3, which is a single 3D 17x17 assembly at BOL without feedback where the materials are defined using an include statement verain/test/Sample_p3_unit_test.inp None
25	The VERAIn package shall be able to parse Problem 4, which are 9 3D 17x17 assemblies at BOL without feedback verain/test/Sample_p4.inp None
26	The VERAIn package shall be able to parse Problem 4, which are 9 3D 17x17 assemblies at BOL without feedback with numerous MPACT options verain/test/Sample_p4_mpact.inp None
27	The VERAIn package shall be able to parse Problem 4a, which is a single 3D 17x17 assembly with a control rod at BOL without feedback verain/test/Sample_p4a_unit_test.inp None
28	The VERAIn package shall be able to parse Problem 6a, which is a single 3D 17x17 assembly at BOL with feedback verain/test/Sample_p6a.inp None
29	The VERAIn package shall be able to parse a modified Problem 6a, which is a single 3D 17x17assembly at BOL with feedback with more complicated fuel types verain/test/Sample_p6a_multifuel.inp # 2907

Req. ID	*Requirement Description* *Test Name* *Test Input* *Additional Info*
30	The VERAIn package shall be able to parse Problem 7, which is a full core 3D problem with feedback verain/test/Sample_p7.inp None
31	The VERAIn package shall be able to parse Problem 9, which is a full core 3D problem with feedback and depletion, using the jumpin capability verain/test/Sample_p9_jumpin.inp None
32	The VERAIn package shall be able to parse Problem 9, which is a full core 3D problem with feedback and depletion, using the restart capability verain/test/Sample_p9_restart.inp None
33	The VERAIn package shall be able to parse Problem 9, which is a full core 3D problem with feedback and depletion, using the fuel shuffle capability verain/test/Sample_p9_shuffle.inp None
34	The VERAIn package shall be able to parse a 3D core of 3x3 assemblies. verain/test/Sample_qtr_core_wsym_3x3_standalone.inp None
35	The VERAIn package shall be able to parse a 3D Seabrook 5x5 assembly to test DNB runs verain/test/Seabrook_5x5rod_HFP.inp None

Req. ID	*Requirement Description* *Test Name* *Test Input* *Additional Info*
36	The VERAIn package shall be able to parse a 2D APR1000 Initial core in quarter symmetry verain/test/apr1400-cyc1.inp None
37	The VERAIn package shall be able to parse inputs in which a BWR axial_void distribution is specified verain/test/bwr-axialvoid.inp None
38	The VERAIn package shall be able to process the azimuthal XS option verain/test/bwr-azixs.inp # 6432
39	The VERAIn package shall be able to parse a 2D GE9 BWR lattice verain/test/bwr-ge9.inp None
40	The VERAIn package shall be able to parse a singe 3D assembly in which BWR conditions and dimensions are used verain/test/bwr-p6-test.inp None
41	The VERAIn package shall be able to parse a 2D BWR Peach Bottom 2 7x7 assembly with an 80 mil channel in full symmetry with no gad and no water rods verain/test/bwr-peach-1.inp None

Req. ID	*Requirement Description* *Test Name* *Test Input* *Additional Info*
42	The VERAIn package shall be able to parse a 2D BWR Peach Bottom 2 7x7 assembly with an 80 mil channel in full symmetry with 3% gad pins and no water rods verain/test/bwr-peach-2.inp None
43	The VERAIn package shall be able to parse a 2D BWR Peach Bottom 2 7x7 assembly with an 80 mil channel in full symmetry with 3% and 4% gad pins and no water rods verain/test/bwr-peach-3.inp None
44	The VERAIn package shall be able to parse a 2D BWR Peach Bottom 2 8x8 assembly with a 100 mil channel in full symmetry with 3% gad pins and large water rods verain/test/bwr-peach-4.inp None
45	The VERAIn package shall be able to parse a 2D BWR Peach Bottom 2 8x8 assembly with a 100 mil channel in full symmetry with 2% gad pins and large water rods verain/test/bwr-peach-5.inp None
46	The VERAIn package shall be able to parse a 2D BWR Peach Bottom 2 8x8 assembly with a 100 mil channel in full symmetry with 2% gad pins and two large water rods verain/test/bwr-peach-6.inp None
47	The VERAIn package shall be able to parse the default material name 'cool' verain/test/bwr-plr.inp # 6458

Req. ID	*Requirement Description* *Test Name* *Test Input* *Additional Info*
48	The VERAIn package shall be able to parse a 2D BWR Peach Bottom 2 8x8 assembly with 100 mil corners and 80 mil channels in full symmetry with 2% gad pins and large water rods verain/test/bwr-thickthincb.inp None
49	VERAIn shall allow the user to specify an orifice map for BWR models which allows for the definition of inlet form losses and flow areas verain/test/bwr_jagged.inp None
50	The VERAIn package shall be able to parse a single 3D 3x3 assembly with TH feedback enabled verain/test/chen_3x3.inp None
51	The VERAIn package shall be able to parse five 3D 3x3 assemblies with TH feedback enabled arranged in a cross with a baffle verain/test/cross_w_baffle.inp None
52	The VERAIn package shall be able to parse a single 3D 17x17 assembly with TH feedback enabled and CTF convergence criteria specified verain/test/ctf_convergence_terms.inp None
53	The VERAIn package shall throw an error for an input in which the axial heights are listed in descending order; the parser shall appropriately fail and describe the input error verain/test/decreasing-axial.inp None

Req. ID	*Requirement Description* *Test Name* *Test Input* *Additional Info*
54	The VERAIn package shall be able to parse inputs in which the default density has been changed verain/test/default_density_change.inp None
55	The VERAIn package shall be able to parse inputs in which xml2ctf is disabled during VERA init verain/test/disable_xml2ctf.inp None
56	VERAIn shall process BWR blade and detector maps verain/test/ex7_bwr.inp None
57	The VERAIn package shall be able to parse a single 3D 3x3 assembly at HFP. verain/test/fast_3x3rod_HFP.inp None
58	The VERAIn package shall throw an error for an input in which a hyphen is given before the assembly name in the assembly map. The parser shall appropriately fail and describe the input error verain/test/front-dash.inp None
59	The VERAIn package shall be able to parse inputs with complex fuel and material specifications verain/test/fuel_options.inp None

Req. ID	*Requirement Description* *Test Name* *Test Input* *Additional Info*
60	The VERAIn package shall be able to parse inputs in which the COBRATF/crud and mass balance/steam generator cards are specified verain/test/mamba_mass_bal.inp None
61	The VERAIn package shall be able to parse inputs in which numerous miscellaneous options are enabled simultaneously verain/test/misc_options.inp None
62	The VERAIn package shall be able to parse a single MOX PWR assembly with three different plutonium enrichment zones verain/test/mox1.inp # 4378
63	The VERAIn package shall not be able to parse an input in which multiple scalar parameters are specified on the same line. The parser shall appropriately fail and describe the input error verain/test/multi_scalar_error.inp None
64	The VERAIn package shall be able to parse a single 3D 17x17 assembly with feedback enabled with 1,300 ppm boron verain/test/p6a_sa_1300_vuq.inp None
65	The VERAIn package shall be able to parse 5 3D 17x17 assemblies with feedback enabled arranged in a cross with parallel 1 set in the COBRATF block verain/test/par_cross.inp None

Req. ID	*Requirement Description* *Test Name* *Test Input* *Additional Info*
66	The VERAIn package shall be able to parse a single 16x16 CE design PWR assembly with Erbia and two enrichments verain/test/pwr-CE-2a.inp None
67	The VERAIn package shall be able to parse a single 16x16 CE design PWR assembly with Erbia and three enrichments verain/test/pwr-CE-2b.inp None
68	The VERAIn package shall be able to parse a single 16x16 CE design PWR assembly verain/test/pwr_CE_single_16x16.inp None
69	The VERAIn package shall be able to parse a single 3D 3x3 assembly at HFP with a natural blanket verain/test/small_3x3rod_BProd.inp None
70	The VERAIn package shall be able to parse a single 3D 3x3 assembly at HFP with numerous BISON options enabled verain/test/small_3x3rod_Bison.inp None
71	The VERAIn package shall be able to parse a single 3D 3x3 assembly at HFP with numerous BISON options enabled with solve_type=standalone verain/test/small_3x3rod_Bison_standalone.inp None

Req. ID	*Requirement Description* *Test Name* *Test Input* *Additional Info*
72	The VERAIn package shall be able to parse a single 3D 3x3 assembly at HFP. verain/test/small_3x3rod_HFP.inp None
73	The VERAIn package shall be able to parse a single 3D 3x3 assembly at HFP with two fuel enrichments verain/test/small_3x3rod_HFP_2fuel.inp None
74	The VERAIn package shall be able to parse a single 3D 3x3 assembly at 10% power verain/test/small_3x3rod_HFP_8g.inp None
75	The VERAIn package shall be able to parse a single 3D 3x3 assembly at HZP verain/test/small_3x3rod_HZP.inp None
76	The VERAIn package shall be able to parse a single 3D 3x3 assembly at HFP with IFBA pins verain/test/small_3x3rod_IFBA.inp None
77	The VERAIn package shall be able to parse a five 3D 3x3 assemblies at HFP when arranged in a cross verain/test/small_5cross_assem_3x3rod_HFP.inp None

Req. ID	*Requirement Description* *Test Name* *Test Input* *Additional Info*
78	The VERAIn package shall be able to parse a five 3D 3x3 assemblies at 10% power when arranged in a cross with numerous COUPLING block options verain/test/small_multibundle.inp None
79	The VERAIn package shall be able to parse a five 3D 3x3 assemblies at 10% power when arranged in a cross in quarter symmetry with numerous COUPLING block options and when CTF is run in parallel verain/test/small_multibundle_par.inp None
80	The VERAIn package shall be able to parse a full core of 3D 3x3 assemblies at HFP in quarter symmetry verain/test/small_qtr_core_3x3rod_HFP.inp None
81	The VERAIn package shall be able to parse a single 3D fuel pin at HFP in full symmetry verain/test/small_singlerod_HFP.inp None
82	The VERAIn package shall be able to parse a single 3D fuel pin at HFP in full symmetry with materials defined in an include file verain/test/small_singlerod_HFP_8g.inp None
83	The VERAIn package shall be able to parse a single 3D fuel pin at HFP in full symmetry with materials defined in an include file with a boron search enabled verain/test/small_singlerod_HFP_8gb.inp None

Req. ID	*Requirement Description* *Test Name* *Test Input* *Additional Info*
84	The VERAIn package shall be able to parse a full 3D SMR core in quarter symmetry with feedback enabled verain/test/smr-7x7.inp None
85	The VERAIn package shall be able to parse a single 3D fuel pin at HFP in full symmetry with a shortened height verain/test/test-short.inp None
86	The VERAIn package shall be able to parse a 3D 3x3 assembly at HFP in full symmetry with CTF edits enabled verain/test/test_cobra_edits.inp None
87	The VERAIn package shall be able to parse a 3D 3x3 assembly at HFP in full symmetry with CTF edits enabled when material definitions are increased by 100 verain/test/test_cobra_edits_inimat.inp None
88	The VERAIn package shall be able to parse a 3D 17x17 assembly with no feedback in quarter symmetry and with VERASHIFT options enabled verain/test/test_verashift_3a.inp None
89	The VERAIn package shall be able to parse a 3D 17x17 assembly with no feedback in quarter symmetry and with VERASHIFT CADIS options enabled verain/test/test_verashift_3a_cadis.inp None

Req. ID	*Requirement Description* *Test Name* *Test Input* *Additional Info*
90	The VERAIn package shall be able to parse a 3D 17x17 assembly with no feedback in quarter symmetry and with VERASHIFT problem_mode=forward enabled verain/test/test_verashift_3a_forward.inp None
91	The VERAIn package shall be able to parse nine 3D 17x17 assemblies with no feedback in quarter symmetry and with VERASHIFT options enabled verain/test/test_verashift_4mini_nonunique.inp None
92	The VERAIn package shall throw an error for an input modeled in quarter symmetry with VERAShift options enabled and with detectors and a bioshield defined; the vera2omn script shall produce the following—Error: core_translate parameter should match outer vessel radius. Writing Omnibus excore input file Error: detector cannot extend into vessel VERA2OMN failed verain/test/test_verashift_4mini_nonunique_bioshield.inp None
93	The VERAIn package shall be able to parse a 5x5 mini core with no feedback in quarter symmetry and automated ex-core detector and bioshield setup with a vessel; the vera2omn script should be triggered and should produce the proper ex-core geometry input file verain/test/test_verashift_small_core_bioshield.inp None
94	The VERAIn package shall throw an error for an input modeled in quarter symmetry and with automated ex-core detector and bioshield without a vessel definition; the vera2omn script should be triggered and the following error message produced—Error: vera_pressure_vessel is False but automated excore generator will only define the geometry beyond the vessel. VERA2OMN failed verain/test/test_verashift_small_core_bioshield_no_pv.inp None
95	The VERAIn package shall be able to parse inputs that have dashes in the assembly names within the assm_map verain/test/with-dash.inp None

Bibliography

[1] *Quality Assurance Requirements for Nuclear Facility Applications*. Standard ASME-NQA-1-2008 with NQA-1a-2009 Addenda. Three Park Avenue, New York, NY 10016-5990: American Society of Mechanical Engineers, 2009.